

Report
Of
Data Science Internship at SkillCraft Technology

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



Under the Guidance of

Mr. Ramvath Ganesh

By

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CANDIDATE'S DECLARATION

I, Richa Kumari, do hereby declare that the internship report titled "Data Science Internship at SkillCraft Technology" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: Richa Kumari

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my internship at SkillCraft Technology. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during the course of this internship. I am sincerely grateful to my faculty guide, Mr. Ramvath Ganesh, for his continuous support and direction throughout this work.

Finally, I express my gratitude to my parents and family for their constant support and blessings.

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3. INTERNSHIP COMPLETION CERTIFICATE



Certificate of Completion issued by SkillCraft Technology (ID: SCT/JUL26/4530)

4. Project Description

4.1 Introduction

This report presents the work carried out during a one-month internship undertaken with SkillCraft Technology as a Data Science Intern, from 14 July 2026 to 14 August 2026. The internship was undertaken as part of the practical training component of the B.Tech CSE curriculum and was intended to provide exposure to the data science domain, including data handling, statistical analysis, and predictive modelling techniques.

As the internship documentation provided by the organisation consists of an offer letter and a completion certificate, and does not include a task-wise activity log, this report focuses on the technical domain of the internship (Data Science) and presents an academic project undertaken to consolidate and demonstrate the concepts studied during the internship period. The report is organised into sections covering the organisational context, the technical problem addressed, the project undertaken, the tools and methodology used, and the outcomes achieved, followed by the supporting certificate and bibliography.

4.2 Organization Profile

SkillCraft Technology is the organisation with which the internship was undertaken, as recorded in the offer letter (ID: SCT/JUL26/4530, dated 13 July 2026) and the completion certificate (dated 26 August 2026). The organisation offers structured internship programs to students, including a Data Science track, through which interns are given exposure to practical, hands-on work in the field. Beyond the details stated in the offer letter and certificate, no further organisational information has been used in this report, in keeping with the requirement to avoid unverified claims about the organisation.

4.3 Problem Statement

Academic coursework in Data Science and related subjects of the B.Tech CSE curriculum typically covers statistical and machine learning concepts in a largely theoretical manner, with limited opportunity to apply these concepts to complete, end-to-end data workflows. This creates a gap between classroom learning and the practical skills expected in a data science role, such as cleaning real-world datasets, performing

exploratory analysis, building predictive models, and communicating findings clearly. The internship, and the project developed alongside it, address this gap by applying core data science techniques to a realistic problem: predicting customer churn from a structured dataset and deriving actionable insights from the analysis.

4.4 Project Objectives

- To understand the end-to-end workflow of a typical data science project, from raw data to actionable insight.
- To perform systematic data cleaning and preprocessing on a structured dataset.
- To apply exploratory data analysis (EDA) techniques to identify patterns, trends, and relationships in the data.
- To design and implement a machine learning model for customer churn prediction.
- To evaluate the performance of the model using standard classification metrics.
- To visualise key findings using appropriate charts and summary tables.
- To document the project workflow in a clear, reproducible, and professional manner.

4.5 Scope of the Project

The project is scoped as an academic, self-contained case study rather than a production deployment for SkillCraft Technology or any of its clients. It covers the following areas:

- Acquisition and inspection of a publicly available customer dataset.
- Data cleaning, handling of missing values, and feature preparation.
- Exploratory data analysis to understand customer behaviour and churn patterns.
- Development of a classification model to predict the likelihood of customer churn.
- Evaluation of the model and interpretation of results.
- Presentation of findings through tables and visualisations.

The project does not involve any proprietary data, systems, or clients belonging to SkillCraft Technology, and no claims are made regarding deployment in a production or

business environment. It is intended to be explainable and reproducible for academic evaluation.

4.6 Technologies and Tools Used

Category	Tool / Technology	Reason for Selection
Programming Language	Python	Widely used for data science; extensive ecosystem of analysis and ML libraries.
Data Handling	Pandas, NumPy	Efficient handling of tabular data and numerical computation.
Visualization	Matplotlib, Seaborn	Standard libraries for statistical plots and exploratory visualisation.
Machine Learning	Scikit-learn	Provides accessible, well-documented implementations of standard classification algorithms.
Development Environment	Jupyter Notebook / Google Colab	Supports an interactive, step-by-step workflow suited to exploratory work.
Version Control	Git and GitHub	Enables systematic tracking of code changes and maintenance of a project record.

Table 4.1: Technologies and tools used in the project, with justification

4.7 System Architecture

The project follows a linear data-processing pipeline, typical of a data science workflow, structured as follows:

- Input: Raw dataset in CSV format, containing customer demographic and usage attributes.
- Processing: Data cleaning, handling of missing values, encoding of categorical variables, and feature scaling.
- Core Logic: A classification model (e.g., logistic regression or decision-tree-based classifier) trained to predict churn.
- Evaluation: Model assessment using accuracy, precision, recall, and confusion matrix.
- Output: Visual summaries (charts, tables) presenting churn drivers and model performance.

Figure 4.1 (to be inserted as a diagram) should depict this flow as a horizontal sequence of five labelled boxes — Input, Processing, Core Logic/Model, Evaluation, and Output — connected by arrows, consistent with the pipeline described above.

Figure 4.1: Proposed system architecture — data flow from raw input to output (diagram to be inserted)

4.8 Methodology

- 1. Requirement Understanding: Reviewing the internship domain and defining the academic project scope.
- 2. Data Acquisition: Selecting a suitable publicly available dataset relevant to customer churn.
- 3. Data Cleaning and Preprocessing: Handling missing values, duplicates, and inconsistent entries.
- 4. Exploratory Data Analysis: Examining distributions, correlations, and churn-related patterns.
- 5. Model Development: Training a classification model on the prepared dataset.
- 6. Model Evaluation: Assessing performance using standard classification metrics.
- 7. Documentation: Recording the workflow, code, and findings in a structured, reproducible format.

4.9 Expected Outcomes

The outcomes of the internship and the accompanying project are presented below, categorised by their level of verification:

(a) Verified Internship Information

- Successful completion of a one-month Data Science internship at SkillCraft Technology (14 July 2026 – 14 August 2026), as confirmed by the offer letter and the completion certificate (ID: SCT/JUL26/4530).

(b) Academic / Prototype Outcomes

- A cleaned and validated dataset prepared for analysis, developed as a student exercise.

- A set of exploratory visualisations highlighting patterns related to customer churn.
- A working prototype classification model for churn prediction, evaluated on standard metrics, developed for academic demonstration and not deployed for any organisation.

(c) Proposed / Future Work

- Extending the prototype model with additional feature engineering and comparison across multiple algorithms.
- Packaging the analysis as an interactive dashboard for easier exploration of results.

No claims are made in this report regarding production deployment, real company data or clients, quantified business impact (e.g., cost savings or revenue), or formal task-wise mentor evaluation, as such information was not part of the documentation provided by the organisation.

4.10 Certificates and Communication Proof

The Internship Completion Certificate issued by SkillCraft Technology is attached in Chapter 3 of this report, along with the internship ID (SCT/JUL26/4530) referenced on both the offer letter and the certificate for verification. The offer letter, dated 13 July 2026, confirms the internship role (Data Science Intern), duration (14 July 2026 to 14 August 2026), and terms of engagement.

SkillCraft Technology's documentation for this internship did not include a task-wise evaluation record, grading matrix, or individually linked proof URLs for specific assignments, unlike some other internship platforms. Accordingly, this report does not present any fabricated task list, mentor remarks, or proof links beyond what is contained in the offer letter and certificate. Where academic project work is described in Sections 4.5 to 4.9, it is presented explicitly as student-developed, academic/prototype work rather than as organisation-verified deliverables.

5. Bibliography/References

1. SkillCraft Technology, "Internship Offer Letter," Reference ID: SCT/JUL26/4530, 13 July 2026.

2. SkillCraft Technology, "Certificate of Completion – Data Science Internship," Richa Kumari, Reference ID: SCT/JUL26/4530, 26 August 2026.
3. Python Software Foundation, Python Language Reference, <https://docs.python.org>
4. Pandas Development Team, Pandas Documentation, <https://pandas.pydata.org/docs>
5. Scikit-learn Developers, Scikit-learn User Guide, <https://scikit-learn.org/stable>
6. Matplotlib Development Team, Matplotlib Documentation, <https://matplotlib.org/stable>
7. Seaborn, Seaborn: Statistical Data Visualization, <https://seaborn.pydata.org>